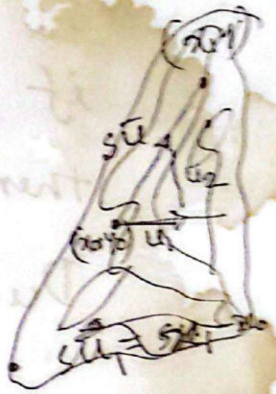
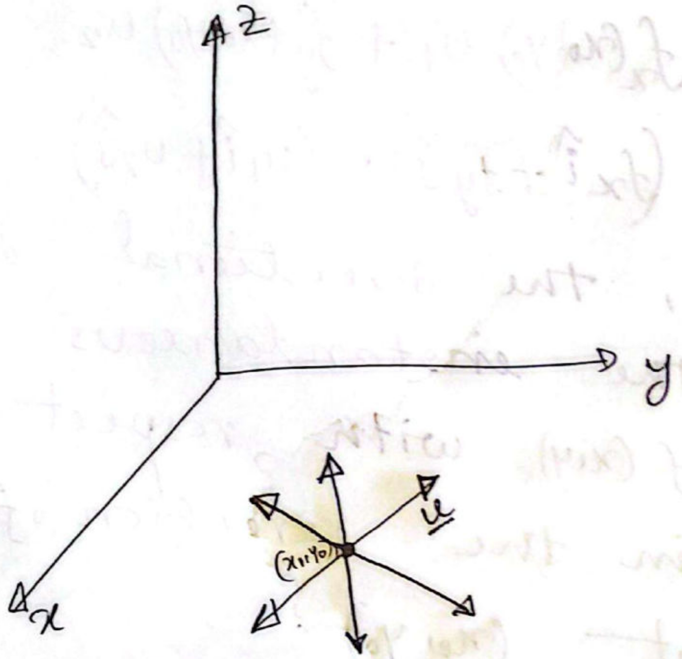
Lec-14

* Directional Derivatives & gradient (13.6 H. Anton)

(Section 13.6 H. Anton)



* Directional derivatives :



Here $\bar{u} = u_1\hat{i} + u_2\hat{j}$

$\|\bar{u}\| = 1$

If $f(x,y)$ is a function of x and y, and if $u = u_1\hat{i} + u_2\hat{j}$ is a unit vector, then the directional derivatives of f in the direction of u at (x_0,y_0) is denoted by $D_u f(x_0,y_0)$ and is defined by

$$D_u f(x_0,y_0) = \frac{d}{ds}\left[f(x_0 + u_1 s,\ y_0 + s u_2)\right]_{s=0}$$

* Theorem: 13.6.3

If $f(x,y)$ is differentiable at (x_0,y_0), & if $\bar{u} = u_1\hat{i} + u_2\hat{j}$ is a unit vector, then the directional derivatives

$$D_u f(x_0,y_0) = f_x(x_0,y_0)\,u_1 + f_y(x_0,y_0)\,u_2$$
$$= \left(f_x\hat{i} + f_y\hat{j}\right)\cdot\left(u_1\hat{i} + u_2\hat{j}\right)$$

* Analytically, the directional derivative represents the instantaneous rate of change of $f(x,y)$ with respect to distance in the direction of u at the point (x_0,y_0).

★ The Gradient: (13.6.4)

(a) If f is a function of x and y, then the gradient of f is defined by

$$\cancel{\nabla f(x,y)} \quad \nabla f(x,y) = f_x(x,y)\,\hat{i} + f_y(x,y)\,\hat{j}$$

(b). If $f(x,y,z)$, then

$$\nabla f(x,y,z) = f_x(x,y,z)\,\hat{i} + f_y(x,y,z)\,\hat{j} + f_z(x,y,z)\,\hat{k}$$

★ One of mathematical meanings of gradient is, the gradient vector determines the orientation of the tangent plan for a particular point.

Theorem- 13.9.3:
(Constrained-Extremum principle for two variables and one constraints)

let f and g be functions of two variables with continuous first partial derivatives on some open set containing the constraint $g(x,y)=0$, and assume that $\nabla g \neq 0$ at any point on this curve. If f has a constrainted relative extrimum, then this extremum occurs at a point (x_0, y_0) on the constraint curve at which the gradient vectors $\nabla f(x_0, y_1)$ and $\nabla g(x_0, y_0)$ are parallel; that is, there is some number λ such that

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0) \rightarrow (1)$$

This scalar is called a lagrange multiplier.

Thus, the method of lagrange multipliers for finding constrained relative extrema is to look for points on the constraint curve $g(x,y)=0$ at which equation (1) is satisfied for some scalar λ.

★ Example: At what point or points on the circle $x^2+y^2=1$ does $f(x,y)=xy$ have an absolute maximum, and what is that maximum?.

Ans: Before Answering the question, we need to know the following theorem.

⊛ Theorem - 13.8.3: (Extreme-value theorem)
If $f(x,y)$ is continuous on a closed and bounded set R, then f has both an absolute maximum and an absolute minimum on R.

The circle $x^2+y^2=1$ is a closed and bounded set and $f(x,y) = xy$ is a continuous function, so ~~A~~ according to Extreme value theorem, $f(x,y)$ has an absolute maximum and an absolute minimum on the circle $x^2+y^2=1$.

To find these extrema, we will use Lagrange multipliers to find the constrained relative extrema.

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and then we will evaluate f at those relative extrema to find the absolute extrema.

A.T.Q., we want to maximize

$f(x,y) = xy$ subject to constraint

$$g(x,y) = x^2+y^2-1 = 0 \longrightarrow (*)$$

According to theorem ~~13.8~~ 13.9.3.

$$\nabla f(x,y) = \lambda \nabla g(x,y) \longrightarrow (1)$$

$\lambda \in \mathbb{R}$.

Here
$$\nabla f(x,y) = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j}\right) f(x,y)$$
$$= \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j}$$
$$= \frac{\partial}{\partial x}(xy)\,\hat{i} + \frac{\partial}{\partial y}(xy)\,\hat{j}$$
$$\nabla f(x,y) = y\hat{i} + x\hat{j} \longrightarrow (2)$$

&
$$\nabla g(x,y) = \left(\frac{\partial}{\partial x} g(x,y)\,\hat{i} + \frac{\partial}{\partial y} g(x,y)\hat{j}\right)$$
$$= \frac{\partial}{\partial x}(x^2+y^2-1)\,\hat{i} + \frac{\partial}{\partial y}(x^2+y^2-1)\,\hat{j}$$
$$\nabla g(x,y) = 2x\hat{i} + 2y\hat{j} \longrightarrow (3)$$

Here,

$$\nabla g(x,y) = \bar{0}$$

$$\Rightarrow 2x\hat{i} + 2y\hat{j} = \bar{0}$$

$$\Rightarrow 2x\hat{i} + 2y\hat{j} = 0\hat{i} + 0\hat{j}$$

$$\Rightarrow 2x = 0 \ ; \ 2y = 0$$

$$\therefore x = 0 \ ; \ y = 0$$

So, the gradient of $g(x,y)$ is zero only at $(0,0)$, but $(0,0)$ is not ~~on~~ bilong to the given circle $x^2+y^2-1=0$

$\Rightarrow \nabla g \neq 0$ at any point on the circle $x^2+y^2=1$.

From (1)

$$\nabla f = \lambda \nabla g$$

$$\Rightarrow y\hat{i} + x\hat{j} = \lambda (2x\hat{i} + 2y\hat{j})$$

$$\Rightarrow y\hat{i} + x\hat{j} = 2x\lambda\hat{i} + 2y\lambda\hat{j}$$

$\Rightarrow y = 2x\lambda$ | & $x = 2y\lambda$

$\Rightarrow \lambda = \dfrac{y}{2x}$ | $\Rightarrow \lambda = \dfrac{\cancel{2y}\, x}{\cancel{x}\, 2y}$

$$\therefore \lambda = \lambda \Rightarrow \frac{y}{2x} = \frac{x}{2y}$$

$\Rightarrow y^2 = x^2$, putting $y^2 = x^2$ in eq. (*)

From (*) $\quad x^2 + y^2 - 1 = 0$

$\Rightarrow x^2 + x^2 - 1 = 0$

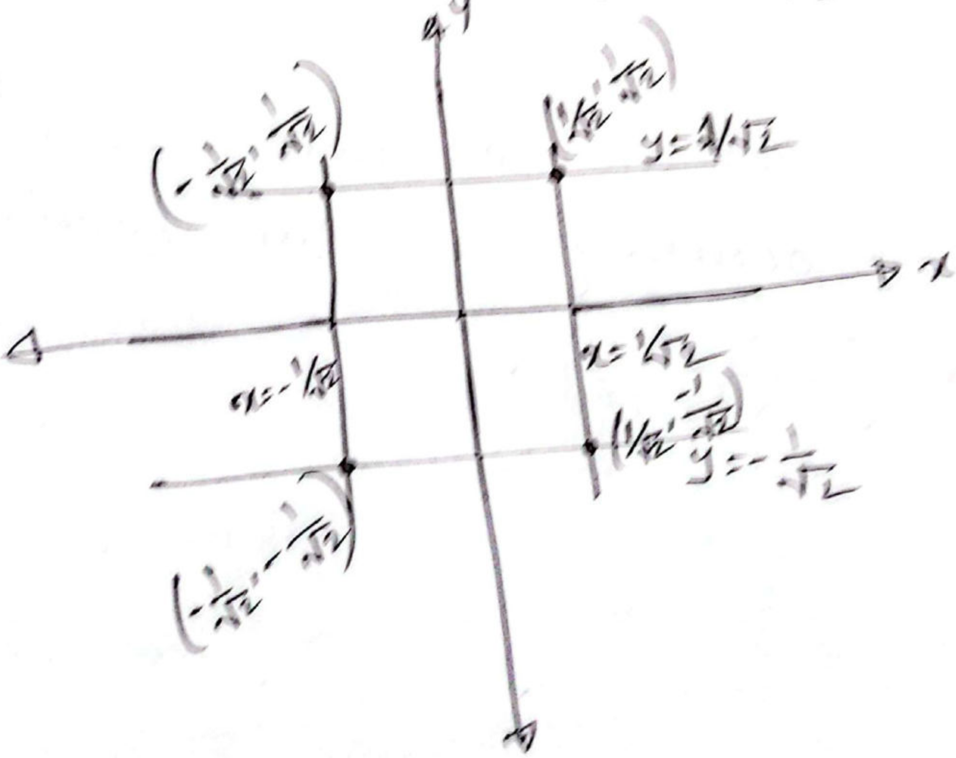
$\Rightarrow 2x^2 = 1$

$\Rightarrow x = \pm \frac{1}{\sqrt{2}}$

putting $x = \pm \frac{1}{\sqrt{2}}$ in $y^2 = x^2$

we get $y^2 = \left(\pm \frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$

$\Rightarrow y = \pm \frac{1}{\sqrt{2}}$



thus, the relative extrema occur at the points $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, $\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ and $\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$.

(x,y)	$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$	$\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$	$\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$	$\left(-\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$	
$f(x,y)$	$1/2$	$-1/2$	$-1/2$	$1/2$	

Thus, the function $f(x,y) = xy$ has an absolute ~~ext~~ maximum of $1/2$ occuring at the two points $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ & $\left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$.

Also, f has an absolute minimum of $-\frac{1}{2}$ occuring at the points $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ and $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

Example 02: Use Lagrange Multipliers to find the maximum and minimum values of $f(x,y,z) = xyz$ subject to the constraint $x^2+y^2+z^2=1$. Also find the points at which these extreme values occur.

Ans: Given $f(x,y,z) = xyz \longrightarrow (1)$

constraint, $g(x,y,z) = x^2+y^2+z^2-1=0 \longrightarrow (2)$

Here the function $f(x,y,z) = xyz$ is continuous in $\mathbb{R}^3$ and constraint $g(x,y,z) = x^2+y^2+z^2-1=0$ is a sphere of unit radius centred at $(0,0,0)$ — that is closed and bounded.

So, According to Extreme value theorem, the function has an absolute maximum & an absolute minimum on $g(x,y,z)=0$.

so; $\nabla f = \frac{\partial}{\partial x} f\hat{i} + \frac{\partial}{\partial y} f\hat{j} + \frac{\partial}{\partial z} f\hat{k}$

$\therefore \nabla f = yz\hat{i} + xz\hat{j} + xy\hat{k} \longrightarrow (3)$

& $\nabla g = \frac{\partial}{\partial x} g\hat{i} + \frac{\partial}{\partial y} g\hat{j} + \frac{\partial}{\partial z} g\hat{k}$

$\therefore \nabla g = 2x\hat{i} + 2y\hat{j} + 2z\hat{k} \longrightarrow (4)$

According to theorem 13.9.3

$\nabla f = \lambda \nabla g \quad ; \in \lambda \in \mathbb{R}$

$\Rightarrow yz\hat{i} + xz\hat{j} + xy\hat{k} = \lambda(2x\hat{i} + 2y\hat{j} + 2z\hat{k})$

$\Rightarrow$ $yz = 2x\lambda$, $\Rightarrow \lambda = \frac{yz}{2x}$ | $xz = 2y\lambda$, $\therefore \lambda = \frac{xz}{2y}$ | $xy = 2z\lambda$, $\therefore \lambda = \frac{xy}{2z}$

$$\therefore \underbrace{\frac{yz}{2x} = \frac{xz}{2y}}_{y^2 = x^2} \ \& \ \underbrace{\frac{xz}{2y} = \frac{xy}{2z}}_{z^2 = x^2} \longrightarrow (5)$$

$\Rightarrow$ $y^2 = x^2$ & $z^2 = x^2$

putting the values of y^2 & z^2 into equation (2)

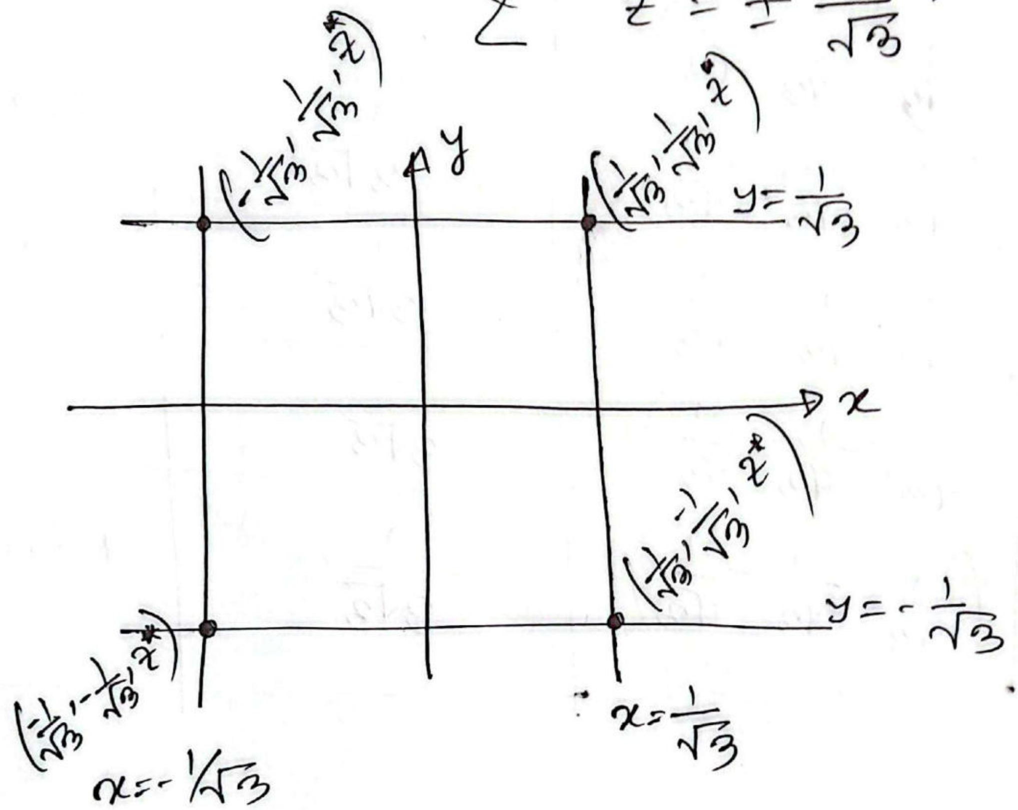
$$x^2+x^2+x^2=1$$

$$\Rightarrow 3x^2=1$$

$\therefore x^* = \pm\frac{1}{\sqrt{3}}$. also, $y^2 = x^2 = \frac{1}{3}$

$$\therefore y^* = \pm\frac{1}{\sqrt{3}}$$

& $z^* = \pm\frac{1}{\sqrt{3}}$.



Total eight possibilities,

$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$; $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$

$\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$; $\left(\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$

$\left(-\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$; $\left(\frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$

$\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$; $\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$

Sl.	point	$f(x,y,z) = xyz$	comment
1	$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$	$\frac{1}{3\sqrt{3}}$	Max
2	$\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$	$-\frac{1}{3\sqrt{3}}$	Min
3	$\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$	$-\frac{1}{3\sqrt{3}}$	Min
4	$\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$	$\frac{1}{3\sqrt{3}}$	Max
5	$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$	$-\frac{1}{3\sqrt{3}}$	Min
6	$\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$	$\frac{1}{3\sqrt{3}}$	Max
7	$\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$	$\frac{1}{3\sqrt{3}}$	Max
8	$\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$	$-\frac{1}{3\sqrt{3}}$	Min

$V = \{(x,y) : x \geq 0\}$ on $\mathbb{R}$

$\mathbb{R}^2 \to \mathbb{R}$

$\int \frac{1}{x^2+y^2}\,dy$ $\quad y^2 = x$

$\mathbb{R}$ $f : A \to B$